VIKRAMA SIMHAPURI UNIVERSITY, NELLORE
DEPARTMENT OF COMPUTER SCIENCE

Syllabus for Master of Computer Applications (2 Year Course) for V.S. University Constituent Colleges(S) and Affiliated Colleges under the jurisdiction of Vikrama Simhapuri University, Nellore with effect from the Academic Year 2025-"26

Programme	**MCA**	**Semester**	**Third**		
Course Code	301	**Course Name**	**Software project Management**		
Course Category	Core course-7	Hours/Week	L	T	P
			3	1	0
		Credits	4		
Course Objectives	1. To understand project management basics. 2. To plan and schedule software projects. 3. To manage project resources and teams. 4. To monitor project progress and risks 5. To deliver quality software on time and budget.				
Course Outcomes	1. Outline the features of different life cycle models. 2. Explain the principles involved in gathering and validating software requirements. 3. Make use of suitable models through analysis of requirements and arrive atan appropriate software design. 4. Appreciate the quality assurance procedures during software development				
Unit-1	**PROJECT EVALUATION AND PROJECT PLANNING** Importance of Software Project Management – Activities Methodologies – Categorization of Software Projects – Setting objectives – Management Principles – Management Control – Project portfolio Management – Cost-benefit evaluation technology – Risk evaluation – Strategic program Management – Stepwise Project Planning				
Unit-2	**PROJECT LIFE CYCLE AND EFFORT ESTIMATION** Software process and Process Models – Choice of Process models - mental delivery – Rapid Application development – Agile methods – Extreme Programming – SCRUM – Managing interactive processes – Basics of Software estimation – Effort and Cost estimation techniques – COSMIC Full function points - COCOMO II A Parametric Productivity Model - Staffing Pattern				
Unit-3	**ACTIVITY PLANNING AND RISK MANAGEMENT** Objectives of Activity planning – Project schedules – Activities – Sequencing and scheduling –Network Planning models – Forward Pass & Backward Pass techniques – Critical path (CRM) method– Risk identification – Assessment – Monitoring – PERT technique – Monte Carlo simulation –Resource Allocation – Creation of critical patterns – Cost schedules.				

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Unit-4	**PROJECT MANAGEMENT AND CONTROL** Framework for Management and control – Collection of data Project termination – Visualizing progress – Cost monitoring – Earned Value Analysis- Project tracking – Change control- Software Configuration Management – Managing contracts – Contract Management. **Staffing In Software Projects** : Managing people – Organizational behavior – Best methods of staff selection – Motivation – The Oldham-Hackman job characteristic model – Ethical and Programmed concerns – Working in teams – Decision making – Team structures – Virtual teams – Communications genres – Communication plans.
Text Books	1. Software Project Management, Bob Hughes, Mike Cotterell, Rajib Mall, McGraw Hill. 2. Software Engineering: A Practitioner's Approach, Roger S. Pressman & Bruce R. Maxim, McGraw-Hill 3. Software Engineering, Ian Sommerville, Pearson. 4. Applied Software Project Management,Andrew Stellman & Jennifer Greene,O'Reilly Media
References	1. Project Management: A Systems Approach to Planning, Scheduling, and Controlling, Harold Kerzner, Wiley. 2. Agile Project Management with Scrum, Ken Schwaber, Microsoft Press. 3. Peopleware: Productive Projects and Teams, Tom DeMarco & Timothy Lister, Addison-Wesley 4. The Mythical Man-Month, Frederick P. Brooks, Addison-Wesley

CO-PO MAPPING

CO'S	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	-	-	-	2	2	-	--	-	-	-	-	-	-	2	-
CO3	2	-	2	-	2	-	-	-	-	-	-	-	-	-	2
CO4	2	-	-	-	2	-	-	-	-	-	-	-	3	-	-

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Programme	**MCA**	**Semester**	**Third**		
Course Code	**302 (A)**	**Course Name**	**MEAN Stack Development**		
Course Category	Core course-8	Hours/Week	L	T	P
			3	1	0
		Credits	3		
Course Objectives	From the course the student will learn 1. Translate user requirements into the overall architecture and implementation of new systems and Manage Project and coordinate with the Client. 2. Monitor the performance of web applications & infrastructure and Troubleshooting web application with a fast and accurate a resolution 3. Design and implementation of Robust and Scalable Front End Applications.				
Course Outcomes	1. Students will be able to understand the Translate user requirements into the overall architecture and implementation of new systems and Manage Project and coordinate with the Client. 2. Students will be able to Implement intermediate and advanced level web development practices 3. Students will be able to Develop and fully functional website and deploy a web server 4. Students will be able to perform web applications & infrastructure and Troubleshooting web application with a fast and accurate a resolution. 5. Students will be able to Design and implementation of Robust and Scalable Front End Application				
Unit-1	**Node.js:** Introduction, Advantages, Node.js Process Model, Node Package Manager, Node JS Modules, Asynchronous Programming, Callbacks, Node.js Event Loop, Streams and Buffers Connecting Node.js to Database, Web Sockets. **Express.js:** Introduction to Express Framework, Express Routing, MVC Structure and Modules, ApplyingMiddleware, Different Template Engines, Error Handling , REST API , Debugging, Different Template Engines, Process managers for Express apps, Security.				
Unit-2	**Angular.js:** Angular Benefits, Dynamic Binding, AngularJSModules, Controllers,Scope, Views, Alternatives toCustom Directives, Event Directives, Expressions, Built-in and Custom filters, Understanding the Basic Options, Form validationsand States, AngularJS service, Factories, Creating Our Own AngularJS Service, Routing Using ngRoute, Redirects.				
Unit-3	**RESTful Web Services:** Using the Uniform Interface, Designing URIs, Web Linking, and Conditional Requests. **ReactJs:**Welcome to React, A Strong Foundation, React's Past and Future, Learning React: Second Edition Changes, Working with the Files, Pure Function, Page Setup, React DOM, React Elements, ReactDOM, Children, Constructing Elements with Data, React Components, Server Rendering React, Refactoring for Advanced Reusability.				

Unit-4	**MongoDB:** Introduction, Architecture, Features, Simple Examples, Create a Database& Collection in Mongo DB. Application development in MongoDB: Document-oriented data, Deployment and administration.
Text Books	1. Pro Mean Stack Development, ELadElrom, Apress. 2. Restful Web Services Cookbook, Subbu Allamraju, O'Reilly 3. Express.JS Guide,The Comprehensive Book on Express.js, Azat Mardan, Lean Publishing. 4. ANGULARS JS, Gajanan A.Deshmukh, Archana Kothawade, Nirali Prakashan publishers, 2021. 5. Learning React: Modern Patterns for Developing React Apps,Alex Banks & Eve Porcello, second edition, O'Reilly, 2020. 6. MongoDB in Action, Kyle Banker Peter Bakkum Shaun Verch Douglas Garrett Tim Hawkins, Second Edition, 2016
References	1. React: Up & Running: Building Web Applications – by Stoyan Stefanov, O'Riely publications. 2. Angular: Up and Running: Learning Angular, Step by Step, Shyam Seshadri, O'Riely publications.

CO-PO MAPPING

CO'S	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	-	3	-	-	-	-	--	-	-	-	-	-	2	3	-
CO3	-	3	3	-	-	-	-	-	-	-	-	-	-	2	-
CO4	-	3	3	-	-	-	-	-	-	-	-	-	-	-	3
CO5	-	-	3	-	-	-	-	-	-	-	-	-	3	-	2

Mean Stack Development Lab

1. Create a User Interface using Angular Js.
2. Create a Tool Bar with Buttons that change the background of Document.
3. Design a Form using Angular JS
4. Develop an application to establish communication between Client and Server.
5. Implementation of Structural Directive (ng-if) using Angular JS.
6. Write a program to connect to MONGODB.
7. Design an Angular.js application to design and using forms in Mean Stack.
8. Design an application which stores data into database.

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Programme	**MCA**	**Semester**	**Third**		
Course Code	**302 (B)**	**Course Name**	**Internet of Things**		
Course Category	Core Course - 8	Hours/Week	L	T	P
			3	1	0
		Credits	3		
Course Objectives	1. To introduce the IoT terminology, technology and its applications. 2. To explain the concept of sensor networks. 3. To understand wireless technology in IoT devices. 4. To understand various sensor & Actuators with Arduino.				
Course Outcomes	1. Recollect different categories of networks. 2. Understand characteristics, Architectural design and IoT. 3. List of various applications of IoT. 4. Design IoT solutions for a need.				
Unit-1	FUNDAMENTALSOF IOT: Introduction, Definitions & Characteristics of IoT, IoT Architectures, Physical & Logical Design of IoT, Enabling Technologies in IoT, History of IoT, About Things in IoT, The Identifiers in IoT, About the Internet in IoT, IoT frameworks, IoT and M2M. **Applications OF IOT**: Home Automation, Smart Cities, Energy, Retail Management, Logistics, Agriculture, Health and Lifestyle, Industrial IoT, Legal challenges, IoT design Ethics, IoT in Environmental protection.				
Unit-2	**SENSORS NETWORKS**: Definition, Types of sensors, Types of Actuators, Examples and Working, IoT Development Boards: Arduino IDE and Board Types, Raspberripi Development Kit, RFID Principles and Components, Wireless Sensor Networks: History and Context, The node, Connecting nodes, Networking Nodes, WSN and IoT.				
Unit-3	WIRELESS TECHNOLOGIES FOR IOT: WPAN Technologies for IoT: IEEE 802.15.4, Zigbee, HaRT, NFC, Z-Wave, BLE, Bacnet, Modbus. **IP BASED PROTOCOLS FOR IOT**: IPv6, 6LowPAN, LoRA, RPL, REST, AMPQ, CoAP, MQTT. Edge connectivity and protocols. **ARDUINO SIMULATION ENVIRONMENT**: Arduino Uno Architecture, setting up the IDE, Writing arduino Software, Arduino Libraries, Basics of Embedded C Programming for Arduino, Interfacing LED, push button and buzzer with Arduino, Interfacing Arduino with LCD.				
Unit-4	**SENSOR & ACTUATORS WITH ARDUINO**: Overview of Sensors working, Analog and Digital Sensors, Interfacing of Temperature, Humidity, Motion, Light and Gas Sensors with Arduino, Interfacing of Actuators with Arduino, Interfacing of Relay Switch and Servo Motor with Arduino. **DEVELOPING IOT'S**: Implementation of IoT with Arduino, connecting and using various IoT Cloud Based Platforms such as Blynk, Thingspeak, AWS IoT, Google Cloud IoT Core etc. Cloud Computing Fog Computing, Privacy and Security Issues in IoT.				

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Test Books	1. Internet of Things – A Hands-no Approach, Arshdeep Bahga and Vijay Madisetti, Universities Press, 2015, ISBN: 9788173719547. 2. Vijay Madisetti and Arshdeep Bahga, "Internet of Things (A Hands-no Approach)",1st Edition, VPT, 2014. 3. Daniel Minoli, – "Building the Internet of Thing with IPv6 and MIPv6: The Evolving World of M2M Communications", ISBN: 978-1-118-47347-4, Willy Publications. 4. Pethuru Raj and Anupama C. Raman, "The Internet of Things: Enabling Technologies, Platforms, and use Cases", CRC Press
References	1. The Complete reference java 2(fourth edition) by – patrick naughton & herbet schildt(TMH). 2. Object Oriented Programming through JAVA2 by – Thamus WU (Mc Graw Hill).] 3. JAVA 2- Dietel & Dietel (Pearson Edition)

CO-PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	3	3	-	-	-	-	--	-	-	-	-	-	2	-	-
CO3	2	2	-	-	--	-	-	-	-	-	-	-	-	2	-
CO4	-	3	2	-	-	-	-	-	-	-	-	-	2	-	1

Internet of Things Lab

1. Write a program for AnalogReadSerial using Arduino.
2. Write a program for DigitalReadSerial using Arduino.
3. Write a program for Blinking LED using Arduino.
4. Write a program for clearing EEPROM using Arduino.
5. Write a program for calculating water level with water sensor using Arduino.
6. Write a program for rotating the servo motor using Arduino.
7. Write a program for Demonstration of **RFID**: Reading RFID tags using Arduino.
8. Write a program to Setup and basic Python-based GPIO access.

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Programme	MCA	**Semester**	**Third**		
Course Code	303 (A)	**Course Name**	**Digital Image Processing**		
Course Category	Core course-9	Hours/Week	L	T	P
			3	1	0
		Credits	3		
Course Objectives	1. To learn the basic concepts of digital image processing and various image transforms. 2. To familiarize the student with the image enhancement techniques. 3. To expose the student to a broad range of image processing techniques and their applications. 4. To appreciate the use of current technologies that is specific to image processing systems. 5. To expose the students to real-world applications of image processing.				
Course Outcomes	1. Review the fundamental concepts of a digital image processing system. 2. Analyze images in the frequency domain using various transforms. 3. Evaluate the techniques for image enhancement and image restoration. 4. Categorize various compression techniques. 5. Interpret Image compression standards and image segmentation and representation techniques.				
UNIT -1	**Introduction to Digital Image Processing:** Introduction, Applications of Image Processing Steps in Image Processing Applications, Digital Imaging System, Sampling and Quantization, Pixel Connectivity, Distance Measures, Color Fundamentals and Models, File Formats, Image Operations.				
UNIT -2	Image Enhancement: Image Transforms: Discrete Fourier Transform, Fast Fourier Transform, Discrete Cosine Transform, Image Enhancement in Spatial and Frequency Domain, Grey Level Transformations, Histogram Processing, Spatial Filtering, Smoothing And Sharpening, Frequency Domain: Filtering in Frequency Domain.				
UNIT -3	**Image Restoration and Multi-Restoration Analysis:** Multi Resolution Analysis: Image Pyramids, Multi Resolution Expansion, Wavelet Transforms, Image Restoration, Image Degradation Model, Noise Modeling, Blur, Order Statistic Filters, Image Restoration Algorithms. **Color Image processing:** Color models, Converting RGB to HSI and vice-versa, Pseudo color processing, Full-color image processing, Color transformations, Smoothing and sharpening. **Image Compression:** Need for data compression, Different types of compression, Variable length coding-Huffman Coding, Run Length Encoding, Shift codes, Arithmetic coding, Vector Quantization, Lossy Compression: Transform coding, Wavelet coding, Basics of Image compression standards: JPEG, MPEG standards.				
UNIT -4	**Image Segmentation and Feature Extraction:** Image Segmentation Detection of Discontinuities, Edge Operators, Edge Linking and Boundary Detection, Thresholding, Region based Segmentation , Image Features and Extraction, Image Features, Types of Features, Feature Extraction, SIFT, SURF and Texture, Feature Reduction Algorithms. **Image Processing Applications:** Image Classifiers, Supervised Learning, Support Vector Machines, Image Clustering, Unsupervised Learning, Hierarchical and Partition based Clustering Algorithms, EM Algorithm.				

Text Books	1. Digital Image Processing, Rafael C Gonzalez, Richard E Woods, 4rth Edition, Pearson Education, 2018. 2. M.Sonka, V. Hlavac, R.Boyle, Image processing Analysis and Machine Vision Thomson Learning. 3. Digital Image Processing and Analysis, B. Chanda & D. D. Majumder, PHI, 2nd Edition, 2011. 4. 4. S. Sridhar, "Digital Image Processing", Second Edition, Oxford University Press, 2016.
References	1. Fundamentals of Digital Image Processing, Anil K. Jain, PHI, 1994 2. Digital Image Processing, B. Jähne, 6th Edition, Springer India, 2005. 3. Pratt. W.K.,Digital Image Processing, 3rd Edition, John Wiley & Sons. 4. Rosenfled A. & Kak, A.C, 1982, Digital Picture Processing, vol .I & II, Academic Press.

CO-PO MAPPING

CO'S	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO2	2	2	3	-	-	-	-	-	-	-	-	-	2	-	-
CO3	2	3	2	-	-	-	-	-	-	-	-	-	-	-	2
CO4	2	3	-	-	2	-	-	-	-	-	-	-	-	2	-
CO5	-	2	-	2	2	-	-	-	-	-	-	-	2	3	-

Digital Image Processing Lab

1. Develop an application convert color image to grey scale image.
2. Develop an application to Quantize the image to reduce bit depth.
3. Develop an application to perform basic operations: negative, mirror, rotate.
4. Develop an application to Plot histogram before and after histogram equalization.
5. Develop an application to Implement RLE and Huffman encoding n binary/gray images.
6. Develop an application to Apply Discrete Cosine Transform to compress an image.
7. Develop an application to Image Edge detection, line detection and point detection.
8. Develop an application to Image compression techniques

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<table>
<tr><td>Programme</td><td>MCA</td><td>Semester</td><td colspan="3">Third</td></tr>
<tr><td>Course Code</td><td>303 (B)</td><td>Course Name</td><td colspan="3">Machine Learning</td></tr>
<tr><td rowspan="3">Course Category</td><td rowspan="3">Core course-9</td><td rowspan="2">Hours/Week</td><td>L</td><td>T</td><td>P</td></tr>
<tr><td>3</td><td>1</td><td>0</td></tr>
<tr><td>Credits</td><td colspan="3">3</td></tr>
<tr><td>Course Objectives</td><td colspan="5">1. To understand the basic concepts of machine learning and probability theory.
2. To appreciate supervised learning and their applications.
3. To understand unsupervised learning like clustering and EM algorithms.
4. To understand the theoretical and practical aspects of probabilistic graphical models
5. To learn other learning aspects such as reinforcement learning, representation learning, deep learning, neural networks and other technologies.</td></tr>
<tr><td>Course Outcomes</td><td colspan="5">1. Understand the features of machine learning to apply on real world problems
2. Characterize the machine learning algorithms as supervised learning and unsupervised learning and Apply and analyze the various algorithms of supervised and unsupervised learning
3. Analyze the concept of neural networks for learning linear and non-linear activation functions
4. Learn the concepts in Bayesian analysis from probability models and methods
5. Understand the fundamental concepts of Genetic Algorithm and Analyze and design the genetic algorithms for optimization engineering problems</td></tr>
<tr><td>Unit-1</td><td colspan="5">Introduction: Machine Learning, Types of Machine Learning, Supervised Learning, Unsupervised Learning, Basic Concepts in Machine Learning, Machine Learning Process, Weight Space , Testing Machine Learning Algorithms , A Brief Review of Probability Theory, Turning Data into Probabilities , The Bias-Variance Tradeoff.
Overview and Introduction to Bayes Decision Theory: Machine intelligence and applications, pattern recognition concepts classification, regression, feature selection, supervised learning class conditional probability distributions, Examples of classifiers bayes optimal classifier and error, learning classification approaches.</td></tr>
<tr><td>Unit-2</td><td colspan="5">Supervised Learning: Linear Models for Regression, Linear Basis Function Models, The Bias-Variance Decomposition, Bayesian Linear Regression, Common Regression Algorithms , Simple Linear Regression, Multiple Linear Regression, Linear Models for Classification, Discriminant Functions , Probabilistic Generative Models, Probabilistic Discriminative Models , Laplace Approximation , Bayesian Logistic Regression , Common Classification Algorithms , k-Nearest Neighbors , Decision trees, Random Forest model, Support Vector Machines.</td></tr>
<tr><td>Unit-3</td><td colspan="5">Unsupervised Learning: Mixture Models and EM, K-Means Clustering, Dirichlet Process Mixture Models, Spectral Clustering, Hierarchical Clustering, The Curse of Dimensionality, Dimensionality, Reduction, Principal Component Analysis, Latent Variable Models(LVM), Latent Dirichlet Allocation (LDA).</td></tr>
<tr><td>Unit-4</td><td colspan="5">Bayesian Networks: Conditional Independence, Markov Random Fields, Learning, Naive Bayes Classifiers, Markov Model, Hidden Markov Model.
Advanced Learning: Reinforcement Learning, Representation Learning, Neural Networks, Active Learning, Ensemble Learning, Bootstrap Aggregation, Boosting, Gradient Boosting Machines, Deep Learning.</td></tr>
<tr><td>Text Books</td><td colspan="5">1. Ethem Alpaydin, “Introduction to Machine Learning” , Third Edition, Prentice Hall of India, 2015.
2. Christopher Bishop, “Pattern Recognition and Machine Learning”, Springer, 2006.
3. Kevin P. Murphy, “Machine Learning: A Probabilistic Perspective”, MIT Press, 2012.
4. 4. Stephen Marsland, “Machine Learning – An Algorithmic Perspective”, Second Edition, CRC Press, 2014.</td></tr>
</table>

References	1. Tom Mitchell, "Machine Learning", McGraw-Hill, 2017. 2. Trevor Hastie, Robert Tibshirani, Jerome Friedman, "The Elements of Statistical Learning", Second Edition, Springer, 2008. 3. Fabio Nelli, "Python Data Analytics with Pandas, Numpy, and Matplotlib", Second Edition, Apress, 2018.

CO-PO MAPPING

CO'S	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-	2
CO2	-	3	-	2	2	-	–	-	-	-	-	-	3	-	-
CO3	-	3	-	-	2	-	-	-	-	-	-	-	-	2	-
CO4	-	3	2	-	2	-	-	-	-	-	-	-	-	2	-
CO5	3	-	-	-	3	-	-	-	-	-	-	-	2	-	2

MACHINE LEARNING LAB

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
2. For a given set of training data examples stored in a .CSV file.
3. Implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.
4. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.
5. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set.
6. Write a program to implement K-Means Clustering.
7. Write a program to implement Bootstrap Aggregation in machine learning.
8. Write a simple machine learning program to implement deep learning.

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<table>
<tr><td>Programme</td><td>MCA</td><td>Semester</td><td colspan="3">Third</td></tr>
<tr><td>Course Code</td><td>305 (A)</td><td>Course Name</td><td colspan="3">MOBILE APPLICATION DEVELOPMENT</td></tr>
<tr><td rowspan="3">Course Category</td><td rowspan="3">Skill oriented Course-5</td><td rowspan="2">Hours/Week</td><td>L</td><td>T</td><td>P</td></tr>
<tr><td>3</td><td>1</td><td>0</td></tr>
<tr><td>Credits</td><td colspan="3">3</td></tr>
<tr><td>Course Objectives</td><td colspan="5">1. To understand the mobile platform and development tools.
2. To design and build user friendly mobile interfaces.
3. To develop functional mobile application.
4. To integrate web services and ensure app security.
5. To test, deploy and maintain mobile applications.</td></tr>
<tr><td>Course Outcomes</td><td colspan="5">1. Analyzearchitectureofandroidandcurrenttrendsinmobileoperatingsystems.
2. Apply suitable software tools and APIs for the development User Interface of a particular Mobile application.
3. Apply intents and broadcast receivers in android application.
4. Develop and design apps for mobile devices using SQLite Database.</td></tr>
<tr><td>Unit-1</td><td colspan="5">Introduction to Android Operating System: AndroidOS and Features–Android development framework; Installing and running applications on Android Studio, Creating AVDs, Types of Android application; Creating Activities, Activity LifeCycle, Activity states, monitoring state changes.</td></tr>
<tr><td>Unit-2</td><td colspan="5">Android application components–Android Manifest file, Externalizingre courses like Simple Values, Drawables, Layouts, Menus, etc,
Building User Interfaces: Fundamental AndroidUI design, Layouts –Linear, Relative, Grid and Table Layouts. User Interface (UI) Components</td></tr>
<tr><td>Unit-3</td><td colspan="5">Fragments–Creating fragments, Life cycle off fragments, Fragment states, Adding fragments to Activity, adding, removing and replacing fragments with fragment transactions, interfacing between fragments and Activities.</td></tr>
<tr><td>Unit-4</td><td colspan="5">Intents and Broadcasts: Using intents to launch Activities, Types of Intents, Passing data to Intents, Getting results from Activities, Broadcast Receivers –Using Intent filters to service implicit Intents, Resolving Intent filters;

Database: Introduction to SQLite database, creating and opening a database, creating tables, inserting retrieving and deleting data;</td></tr>
<tr><td>Text Books</td><td colspan="5">1. Professional Android4 Application Development, RetoMeier, Wiley India,(Wrox),2012
2. Android Application Development for Java Programmers, James C Sheusi, Cengage Learning, 2013</td></tr>
</table>

References	1. BeginningAndroid4ApplicationDevelopment,Wei MengLee, Wiley India (Wrox),2013 2. AndroidApplicationDevelopment(withKitkatSupport),BlackBook,PradeepKothari,2014, Dreamtech Press publisher, Kogent Learning Inc., 2014 3. AndroidProgramming:PushingtheLimits,ErikHellman,1st Edition,Wiley Publications, 201

CO –PO MAPPING

CO'S	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	2	-	-	--	-	-	-	-	-	3	-	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO5	-	-	-	-	3	-	-	-	-	-	-	-	-	-	3

MOBILE APPLICATION DEVELOPMENT LAB

1. Develop a mobile application on android to design a Login Screen.
2. Develop a mobile application to design a basic calculator.
3. Develop an application that uses Layout Managers and event listeners.
4. Develop an application that makes use of Notification Manager.
5. Develop an application to design a STOP watch.
6. Develop a mobile application to send an email.
7. Develop an application that uses GUI components, Font and Colours.
8. Implement an application that uses Multi-threading.

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<table>
<tr><td>Programme</td><td>MCA</td><td>Semester</td><td colspan="3">Third</td></tr>
<tr><td>Course Code</td><td>305 (B)</td><td>Course Name</td><td colspan="3">Web interface Design</td></tr>
<tr><td rowspan="3">Course Category</td><td rowspan="3">Skill oriented Course-5</td><td rowspan="2">Hours/Week</td><td>L</td><td>T</td><td>P</td></tr>
<tr><td>3</td><td>1</td><td>0</td></tr>
<tr><td>Credits</td><td colspan="3">3</td></tr>
<tr><td>Course Objectives</td><td colspan="5">From the course the student will learn
1. To learn basic UI/UX concepts
2. To design responsive web pages
3. To learn using HTML, CSS, and JavaScripts
4. To learn how to apply visual design principles</td></tr>
<tr><td>Course Outcomes</td><td colspan="5">1. Apply core design principles to create intuitive and user-friendly web interfaces.
2. Build responsive and accessible web layouts using HTML, CSS, and Bootstrap.
3. Design interactive prototypes using modern UI tools like Figma or Adobe XD.
4. Evaluate web interfaces using usability testing techniques.
5. Develop a professional-level web interface project suitable for a design portfolio.</td></tr>
<tr><td>Unit-1</td><td colspan="5">Introduction to Web Interface Design, Differences: UI vs UX, Human-Computer Interaction Basics, Principles of Good Interface Design, User-Centered Design (UCD) Process, Information Architecture & Content Strategy, Wireframing Concepts and Low-Fidelity Design</td></tr>
<tr><td>Unit-2</td><td colspan="5">HTML5 Semantic Structure: Header, Section, Article, Footer. CSS3 Styling and Positioning Techniques, Grid and Flexbox Layouts, Responsive Design Principles, Introduction to UI Frameworks: Bootstrap, Design Accessibility Standards,</td></tr>
<tr><td>Unit-3</td><td colspan="5">Typography, Color Theory, and Visual Hierarchy. UI Design Tool Figma, Creating Design Systems and Style Guides, Micro interactions and Feedback, Interactive Elements (Buttons, Forms, Modals), Design for Mobile vs Desktop, Form Design and Validation UX</td></tr>
<tr><td>Unit-4</td><td colspan="5">Usability Testing Techniques (A/B Testing, Heuristics Evaluation), Analytics Tools, Iterative Design and Feedback Loops, User Personas and Journey Mapping, Presentation of UI/UX, Portfolio and Documentation</td></tr>
<tr><td>Reference Books</td><td colspan="5">Don't Make Me Think by Steve Krug – New Riders Publishing
Learning Web Design by Jennifer Niederst Robbins – O'Reilly Media
Designing Interfaces by Jenifer Tidwell – O'Reilly Media
About Face: The Essentials of Interaction Design by Alan Cooper – Wiley
Supplementary: Figma or Adobe XD official documentation for UI prototyping</td></tr>
</table>

CO-PO MAPPING

CO'S	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-	2
CO2	-	3	-	2	2	-	--	-	-	-	-	-	3	-	-
CO3	-	3	-	-	2	-	-	-	-	-	-	-	-	2	-
CO4	-	3	2	-	2	-	-	-	-	-	-	-	-	2	-
CO5	3	-	-	-	3	-	-	-	-	-	-	-	2	-	2

Web Interface Design - Lab

1. Create a basic wire frame for a personal website
2. Build a multi-section responsive website layout
3. Design an interactive login for min Figma.
4. Prototype a mobile-first web app interface
5. Conduct a usability test on a prototype
6. Complete and present final capstone project.
7. Implement **interactive elements**: buttons, forms, modals, and hover states in HTML/CSS/JS.
8. Create a **User-Centered Design (UCD) process flow** for a sample web application.

Syllabus for Master of Computer Applications (2 Year Course) for V.S. University Constituent Colleges(S) and Affiliated Colleges under the jurisdiction of Vikrama Simhapuri University, Nellore with effect from the Academic Year 2025-"26

Programme	**MCA**	**Semester**	**Third**		
Course Code	**306 (A)**	**Course Name**	. **Data Analysis using Power BI**		
Course Category	Skill oriented Course-6	Hours/Week	L	T	P
			3	1	0
		Credits	3		
Course Objectives	From the course the student will learn To understand power BI tools and user interface. 1. Connect to data sources and prepare data using power query. 2. To create visual reports like charts, tables and dashboards 3. Analyse data with DAX. 4. Publish and share reports using power BI service.				
Course Outcomes	1. Understand the Fundamentals of Business Intelligence and Power BI Ecosystem 2. Perform Data Cleaning, Transformation, and Modeling using Power Query and DAX 3. Develop Interactive Visual Reports and Dashboards 4. Implement Power BI Service Features for Sharing, Collaboration, and Security				
Unit-1	**Understanding the Basics** Introduction to Business Intelligence, Advantages of BI system, Components of BI, Concepts of the star and snowflake schema, Key performance indicator, Functions of Power BI, Power BI components, Power BI environment, Different users of Power BI, Installation of Power BI desktop, Power BI desktop interface **Connect and Shape:** Data connections in Power BI, Connecting to data, Understanding Power Query Editor, Loading customers and products tables, Loading data from MS SQL server				
Unit-2	**Data Transformations:** Data Profiling using Query Editor, Data transformation on text data, Data transformation on Numeric Data, Data transformation on Dates, Utilizing Data Source Parameters. Data Model: Introduction to Data Modeling, Review and Create Relationships, Review the Loaded Tables, Review the Data Model, Combining Queries in Power BI.				
Unit-3	**Data Analysis Expressions:** Introduction to DAX, Calculated Columns and Measures, Calculated columns, Mathematical Functions, Count Functions, Information Functions, Logical Functions, Filter functions, Date and time functions, Variables in DAX expressions **Visualizations in Power BI:** Review the data model, Understanding visualization, Introduction to Power BI reports, Standardizing report development, Creating Multi-Page Reports using Visualizations, Filters and Slicers **Drill Through and Drill Down Reports:** Drill through report in Power BI, Drill down report in Power BI, Creating custom hierarchy				

Unit-4	**Artificial Intelligence in Power BI:** Artificial Intelligence in Power BI, Why AI visuals are used, AI visuals in Power BI desktop, Narrative visual, AI in Query Editor **Power BI Service:** Understanding the Power BI service, Foundational elements of Power BI Service, Publishing reports from the Power BI desktop, Visualizations in Power BI Service Creating a dashboard. **Securing Application:** Row-level security in Power BI, Implementing Row-Level Security, Configure RLS in Power BI service.
Reference	"Mastering Power BI" by Chandraish Sinha, 2nd Edition BPB Publication "Analyzing Data with Power BI and Power Pivot for Excel" by Alberto Ferrari and Marco Russo (Microsoft Press) "The Definitive Guide to DAX" by Marco Russo and Alberto Ferrari "Mastering Microsoft Power BI" by Brett Powell. "Power BI Cookbook" by Brett Powell

CO-PO MAPPING

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CO2	2	2	2	2	-	-	--	-	-	-	-	-	-	2	-
CO3	2	2	2	3	-	-	-	-	-	-	-	-	-	-	2
CO4	3	2	3	3	-	-	-	-	-	-	-	-	2	-	2

Data Analysis using Power BI – Lab

1. Connect and Import Data from Excel, CSV, and SQL Server Sources
2. Perform Data Cleaning (nulls, incorrect formats, duplicates) and Transformation using Power Query
3. Creating and Managing Data Models by establishing relationships between tables in a star schema.
4. Creating Calculated Columns and Measures using DAX
5. Perform YTD, MTD, and previous year comparisons Time Intelligence Calculations in DAX
6. Designing Interactive Dashboards with slicers, KPIs, and charts
7. Implementation of Custom Visuals and Advanced Charting
8. Implementing Row-Level Security (RLS)

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DEPARTMENT OF COMPUTER SCIENCE

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<table>
<tr><td>Programme</td><td>MCA</td><td>Semester</td><td colspan="3">Third</td></tr>
<tr><td>Course Code</td><td>306 (B)</td><td>Course Name</td><td colspan="3">No SQL with Mongo DB</td></tr>
<tr><td rowspan="3">Course Category</td><td rowspan="3">Skill oriented Course-6</td><td rowspan="2">Hours/Week</td><td>L</td><td>T</td><td>P</td></tr>
<tr><td>3</td><td>1</td><td>0</td></tr>
<tr><td>Credits</td><td colspan="3">3</td></tr>
<tr><td>Course Objectives</td><td colspan="5">From the course the student will learn
1. To understand NoSQL basics.
2. To explore mango DB structure.
3. To perform CRUD operations.
4. Write query to search and filter data
5. Working with real time data</td></tr>
<tr><td>Course Outcomes</td><td colspan="5">1. Explain NoSQL concepts and MongoDB's role in modern application development.
2. Design and manage document-based schemas for real-world applications.
3. Write efficient queries and aggregations to analyze and manipulate data.
4. Integrate MongoDB with applications and perform basic administrative tasks.</td></tr>
<tr><td>Unit-1</td><td colspan="5">Introduction to NoSQL: History and Evolution, NoSQL vs RDBMS: Concepts and Differences, Types of NoSQL Databases, MongoDB Architecture, MongoDB Installation, Mongo Shell and MongoDB Compass, Creating Databases and Collections, CRUD Operations: Insert, Find, Update, Delete</td></tr>
<tr><td>Unit-2</td><td colspan="5">Schema Design Best Practices in MongoDB, Embedding vs Referencing Documents, Data Types in MongoDB, Working with Arrays and Subdocuments, Query Operators: $eq, $ne, $gt, $lt, $in, $regex. Sorting, Limiting, and Pagination. Indexing Basics: Single Field and Compound Indexes, Performance Considerations in Querying</td></tr>
<tr><td>Unit-3</td><td colspan="5">Introduction to Aggregation Pipeline, Pipeline Stages: $match, $group, $project, $sort, $unwind, etc. Using Expressions in Aggregations, Data Transformation Use Cases, Using $lookup for Joins, Validation with Schema Validators, Performance Optimization Techniques, Introduction to Transactions</td></tr>
<tr><td>Unit-4</td><td colspan="5">MongoDB with Node.js using Mongoose, Security and Role-Based Access Control, Backup and Restore: mongo dump and mongo restore, Monitoring and Performance Tuning, Overview of Replication and Sharding, Use Cases: Real-world Applications of MongoDB</td></tr>
</table>

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Reference Books	1. MongoDB: The Definitive Guide (3rd Edition) by Shannon Bradshaw, Eoin Brazil, Kristina Chodorow – O'Reilly Media 2. Mastering MongoDB (2nd Edition) by Alex Giamas – Packt Publishing 3. NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence by Pramod J. Sadalage& Martin Fowler – Addison-Wesley 4. MongoDB Official Documentation – https://www.mongodb.com/docs/

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CO2	2	-	3	2	2	-	--	-	-	-	-	-	-	2	3
CO3	2	2	2	3	-	-	-	-	-	-	-	-	-	-	-
CO4	2	-	2	3	2	-	-	-	-	-	-	-	2	-	2

No SQL with Mongo DB-Lab

1. Create your first database and collection in Mongo DB
2. Perform basic CRUD operations using Mongo Shell and Compass
3. Design a schema for an e-commerce application
4. Implement indexing for performance
5. Perform multi-stage transformations
6. Use $ lookup to join two collections
7. Connect Mongo DB to a Node.js app using Mongoose
8. Create a RESTFUL API to perform CRUD operations